

Labwork 1 - Introduction to cryptography

Lecturer: Nguyen Minh Huong, ICT Dept, USTH

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Write a program in Matlab, C++ or python to solve below exercises.

Exercise 1

Given two numbers: $a = 123$, $b = 543$, compute:

1. The binary form of a
2. $aXORb$ in binary form and in decimal form

Exercise 2

Given a set of numbers

$$A = \{105, 110, 116, 114, 111, 50, 99, 114, 121, 112, 116, 111, 123, 119, 51, \\ 108, 99, 48, 109, 51, 95, 116, 48, 95, 106, 48, 117, 114, 110, 51, 121, 125\}$$

find the message hidden behind it. Knowing that each number in the set is the ascii code of a letter in the hidden message.

Exercise 3

Given a set of numbers

$$B = \{99, 100, 126, 120, 101, 56, 105, 120, 115, 122, 126, 101, 113, 59, 121, \\ 126, 57, 122, 85, 58, 100, 85, 96, 58, 127, 120, 100, 57, 115, 119\}$$

find the message hidden behind it. Clues are as below:

1. each number in the set is a xor of a hidden number and the number $k = 10$ (decimal)
2. each hidden number is an ascii code of a letter in the message.

Exercise 4

Based on the provided description of the A5/1 cipher, implement the A5/1 Key generation algorithm. List n the keystream bits, suppose that the values of registers are provided X_0, Y_0, Z_0

Test the program with following register values in decimal:

$$X = (x_0, x_1, \dots, x_{18}) = (513365)_{10}$$

$$Y = (y_0, y_1, \dots, y_{21}) = (3355443)_{10}$$

$$Z = (z_0, z_1, \dots, z_{22}) = (7401712)_{10}$$

$$n = 10$$

Note

Documents are provided in assignments on Google Classroom ””.